STAT 4005 - Time Series

Chapter 3 - ARMA Model

Agenda

Introduction

- ARMA(p, q) model
  - The most common model for **stationary** time series

$$\underbrace{Y_t - \phi_1 Y_{t-1} - \cdots - \phi_p Y_{t-p}}_{\text{Autoregressive (AR)}} \quad = \quad \underbrace{Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q}}_{\text{Moving Average (MA)}}$$

    - Y_t is observation,
    - $Z_t \sim$ WN $(0, \sigma^2)$ is white noise.
  - Equivalently, the ARMA(p, q) model can be written as

$$\phi(B) Y_t = \theta(B) Z_t,$$

where

$$\begin{aligned} \phi(B) &= I - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p, \\ \theta(B) &= I - \theta_1 B - \theta_2 B^2 - \cdots - \theta_q B^q, \end{aligned}$$

are **characteristic polynomials** without common roots, i.e., no x s.t. $\phi(x) = \theta(x) = 0$, and B is the backshift operator.

- ARIMA(p, d, q) model
  The most common model for **non-stationary** time series is

$$\phi(B) \underbrace{(1-B)^d}_{\text{Integrated}} Y_t \quad = \quad \theta(B) Z_t.$$

- If Y_t follows an ARIMA(p, d, q) model,
  then $\triangle^d Y_t = (1-B)^d Y_t$ follows an ARMA(p, q) model, where $(1-B)^d Y_t$ is the d-times differenced series from Y_t.

Differencing

For a time series Y_t,

- When $d = 1$, the first order differenced series is $\triangle Y_t = (1 - B)Y_t = Y_t - Y_{t-1}$.
- When $d = 2$, the second order differenced series is

$$
\begin{aligned}
(1 - B)^2 Y_t &= \triangle^2 Y_t \\
&= \triangle(\triangle Y_t) \\
&= \triangle(Y_t - Y_{t-1}) \\
&= (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2}) \\
&= Y_t - 2Y_{t-1} + Y_{t-2} \\
&= (1 - 2B + B^2)Y_t.
\end{aligned}
$$

- In general, the d-th order differenced series can be expressed as

$$
(1 - B)^d Y_t = \sum_{k=0}^{d} \binom{d}{k} (-B)^k Y_t = \sum_{k=0}^{d} \binom{d}{k} (-1)^k Y_{t-k}.
$$

No common root

- Consider $(1-\psi B)Y_t = (1-\psi B)Z_t$, i.e.,

$$Y_t - \psi Y_{t-1} = Z_t - \psi Z_{t-1}\,, \qquad Z_t \sim WN(0,\sigma^2)\,. \tag{1}$$

- Now, $\phi(B) = \theta(B) = (1-\psi B)$
  $\Rightarrow \phi\left(\frac{1}{\psi}\right) = \theta\left(\frac{1}{\psi}\right) = 0 \Rightarrow \frac{1}{\psi}$ is the common root.
- Apply (1) repeatedly, we have

$$Y_t - Z_t = \psi Y_{t-1} - \psi Z_{t-1} = \psi(Y_{t-1} - Z_{t-1}) = \cdots = \psi^k(Y_{t-k} - Z_{t-k})$$

  for all $k \geq 1$.
- Thus, the only **stationary** solution for (1) is

$$Y_t = Z_t$$

  for all $t \geq 0$.
- **Remark:** $Y_t = Z_t + \psi^t$ is a **non-stationary** solution for (1)

Exercise

Identify the following as specific ARIMA(p, d, q) models

a) $Y_t - 0.5Y_{t-1} + 0.06Y_{t-2} = Z_t + 0.6Z_{t-1} - 0.16Z_{t-2}$

b) $Y_t - 0.8Y_{t-1} - 0.2Y_{t-2} = Z_t + 0.2Z_{t-1}$

c) $Y_t - 0.8Y_{t-1} - 0.2Y_{t-2} = Z_t - 0.2Z_{t-1}$

Solution:

a)

$$\begin{aligned}
Y_t - 0.5Y_{t-1} + 0.06Y_{t-2} &= Z_t + 0.6Z_{t-1} - 0.16Z_{t-2} \\
(1 - 0.3B)(1 - 0.2B)Y_t &= (1 + 0.8B)(1 - 0.2B)Z_t \\
(1 - 0.3B)Y_t &= (1 + 0.8B)Z_t.
\end{aligned}$$

It is an ARIMA(1,0,1) model.

Exercise

b)

$$
\begin{aligned}
Y_t - 0.8Y_{t-1} - 0.2Y_{t-2} &= Z_t + 0.2Z_{t-1} \\
(1-B)(1+0.2B)Y_t &= (1+0.2B)Z_t \\
(1-B)Y_t &= Z_t.
\end{aligned}
$$

It is an ARIMA(0,1,0) model.

c)

$$
\begin{aligned}
Y_t - 0.8Y_{t-1} - 0.2Y_{t-2} &= Z_t - 0.2Z_{t-1} \\
(1+0.2B)(1-B)Y_t &= (1-0.2B)Z_t
\end{aligned}
$$

It is an ARIMA(1,1,1) model.

Moving Average Model

Definition 1 (MA(q) model)

A stochastic process $\{Y_t\}_{t=1,2,\ldots}$ follows an MA(q) model if

$$Y_t = Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q}$$

where $Z_t \sim WN(0, \sigma^2)$.

- Sometimes, we further assume $Z_t \overset{\text{iid}}{\sim} N(0, \sigma^2)$

Moving Average Model: Stationarity

Proposition 2 (Stationarity)

For all q and $(\theta_1, \dots, \theta_q) \in \mathbb{R}^q$, the MA($q$) model $Y_t = Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q}, \quad Z_t \sim WN(0, \sigma^2)$ defines a stationary process.

Proof.

Using the properties of white noise, i.e. $\mathrm{E}(Z_t) = 0$, $\mathrm{Cov}(Z_t, Z_k) = \sigma^2 1_{\{t=k\}}$, we have

1. $\mathrm{E}(Y_t) = 0$
2. $\mathrm{Var}(Y_t) = \left(1 + \theta_1^2 + \cdots + \theta_q^2\right) \sigma^2$
3. $\mathrm{Cov}(Y_t, Y_{t+k}) = \begin{cases} 0 & , \quad |k| > q \\ \sigma^2 \sum_{i=0}^{q-|k|} \theta_i \theta_{i+|k|} & , \quad |k| \le q \, , (\theta_0 = -1) \end{cases}$

$\Rightarrow$ MA model is always stationary.

$\square$

Example 3

Find the ACF of the MA(2) model: $Y_t = Z_t + 0.6Z_{t-1} - 0.16Z_{t-2}$, $Z_t \sim$ W.N.$(0, \sigma^2)$.

$$
\begin{aligned}
\gamma(0) &= (1 + (-0.6)^2 + 0.16^2)\sigma^2 = 1.3856\sigma^2, \\
\gamma(1) &= ((-1)\cdot(-0.6) + (-0.6)\cdot 0.16)\sigma^2 = 0.504\sigma^2, \\
\gamma(2) &= (-1)\cdot 0.16\sigma^2 = -0.16\sigma^2, \\
\gamma(k) &= 0 \text{ for } k \geq 3, \\
\rho(1) &= \tfrac{\gamma(1)}{\gamma(0)} = 0.3637, \\
\rho(2) &= \tfrac{\gamma(2)}{\gamma(0)} = -0.1155, \\
\rho(k) &= 0 \text{ for } k \geq 3, \\
\rho(k) &= \rho(-k) \text{ for } k < 0.
\end{aligned}
$$

Example 4

- Different MA models can have the same ACF:

A. MA(1): $Y_t = Z_t - \theta Z_{t-1}\,, \quad Z_t \sim WN(0,1)\,,$
$\gamma(0) = 1 + \theta^2, \quad \gamma(1) = -\theta$

$$\rho(k) = \begin{cases} 1 & , \quad k = 0 \\ -\frac{\theta}{1+\theta^2} & , \quad |k| = 1 \\ 0 & , \quad \text{otherwise} \end{cases}$$

B. MA(1): $X_t = Z_t - \frac{1}{\theta} Z_{t-1}\,, \quad Z_t \sim WN(0,1)\,,$
$\gamma(0) = 1 + 1/\theta^2, \quad \gamma(1) = -1/\theta$

$$\rho(k) = \begin{cases} 1 & , \quad k = 0 \\ -\frac{\theta}{1+\theta^2} & , \quad |k| = 1 \\ 0 & , \quad \text{otherwise} \end{cases}$$

Moving Average Model - Invertibility

Definition 5 (Invertibility)

A model is **invertible** if the white noise Z_t can be expressed as a linear combination of the past observations Y_t, Y_{t-1},...,i.e.,

$$Z_t = \sum_{k=0}^{\infty} \psi_k Y_{t-k} \,,$$

where $\sum_{k=0}^{\infty} |\psi_k| < \infty$ (so that the summation is well defined).

- Example: MA(1) model: $Y_t = Z_t - \theta Z_{t-1}$

$$\begin{aligned}
Z_t &= Y_t + \theta Z_{t-1} = Y_t + \theta(Y_{t-1} + \theta Z_{t-2}) \\
&= Y_t + \theta Y_{t-1} + \theta^2 Z_{t-2} = Y_t + \theta Y_{t-1} + \theta^2 (Y_{t-2} + \theta Z_{t-3}) \\
&= Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Z_{t-3} \\
&= Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Y_{t-3} + \theta^4 Y_{t-4} + \cdots \\
&= \sum_{k=0}^{\infty} \theta^k Y_{t-k} \,, \qquad \text{(only if } |\theta| < 1\text{)}
\end{aligned}$$

Moving Average Model

- For MA(1) model $Y_t = Z_t - \theta Z_{t-1}$ with $|\theta| < 1$, we have

$$Z_t = Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Y_{t-3} + \theta^4 Y_{t-4} + \cdots \tag{2}$$

- Recall the formula $(1-x)^{-1} = 1 + x + x^2 + \cdots + x^k + \cdots$
  - Proof:
    $(1 - x)(1 + x + x^2 + \cdots)$
    $= (1 + x + x^2 + \cdots) - (x + x^2 + \cdots) = 1$
  - Write $Y_t = Z_t - \theta Z_{t-1} = (1 - \theta B)Z_t$, we "invert" $(1 - \theta B)$ to get

$$\begin{aligned} Z_t &= (1-\theta B)^{-1} Y_t \\ &= (1 + \theta B + \theta^2 B^2 + \theta^3 B^3 + \theta^4 B^4 + \cdots) Y_t \\ &= Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \theta^3 Y_{t-3} + \theta^4 Y_{t-4} + \cdots \end{aligned}$$

  - Same as (2)
- $Z_t = (1 - \theta B)^{-1} Y_t = \sum_{j=0}^{\infty} \theta^j Y_{t-j}$ should be understood as notation rather than mathematical operation.

Moving Average Model - Model Selection

We have seen that $\rho(k) = \begin{cases} 1 & , \quad k = 0 \\ -\frac{\theta}{1+\theta^2} & , \quad |k| = 1 \\ 0 & , \quad \text{otherwise} \end{cases}$ is the ACF of either

A. $Y_t = Z_t - \theta Z_{t-1} = (1 - \theta B) Z_t$
$\Rightarrow Z_t = Y_t + \theta Y_{t-1} + \theta^2 Y_{t-2} + \cdots$

B. $Y_t = Z_t - \frac{1}{\theta} Z_{t-1} = (1 - \frac{1}{\theta} B) Z_t$
$\Rightarrow Z_t = Y_t + \frac{1}{\theta} Y_{t-1} + \frac{1}{\theta^2} Y_{t-2} + \cdots$

- Which model is more preferable?
  - If $|\theta| < 1$, series (B) does not converge,
    $\Rightarrow$ (A) is invertible (more preferable)
    (B) is **NOT** invertible
  - If $|\theta| > 1$, series (A) does not converge,
    $\Rightarrow$ (A) is **NOT** invertible
    (B) is invertible (more preferable)

Invertibility for general MA Model

Theorem 6 (Invertibility)

An MA(q) process $\{Y_t\}$ is invertible if the roots of the equation $\theta(B) = 0$ all lie outside the unit circle

Proof.

Factorizing the characteristic polynomial yields

$$\begin{aligned} Y_t &= Z_t - \theta_1 Z_{t-1} - \cdots - \theta_q Z_{t-q} \\ &= \theta(B) Z_t \\ &= (1 - \overline{\theta}_1 B)(1 - \overline{\theta}_2 B) \cdots (1 - \overline{\theta}_q B) Z_t \end{aligned}$$

- $\frac{1}{\overline{\theta}_k}$s are the roots of the equation $\theta(B) = 0$
- $(1 - \overline{\theta}_k B)$ can be "inverted" to the other side if $|\overline{\theta}_k| < 1$
- $\therefore$ The model is **invertible** if all $|\overline{\theta}_k| < 1$
  - all $|\overline{\theta}_k| < 1 \Leftrightarrow$ all $\frac{1}{|\overline{\theta}_k|} > 1 \Leftrightarrow$ all roots lie outside the unit circle

$\square$

Exercise

Which of the following MA model(s) is(are) invertible?

a) $Y_t = Z_t + 0.6Z_{t-1} - 0.16Z_{t-2}$

b) $Y_t = Z_t - 0.6Z_{t-1} - 0.72Z_{t-2}$

c) $Y_t = Z_t - 0.5Z_{t-2} - 0.5Z_{t-4}$

Solution:

a) The roots of the quadratic equation $\theta(x) = 1 + 0.6x - 0.16x^2$ are $x_1 = -1.25$ and $x_2 = 5$. Since $|x_i| > 1$ for $i = 1, 2$, the model is invertible.

b) The roots of the quadratic equation $\theta(x) = 1 - 0.6x - 0.72x^2$ are $x_1 = -1.6667$ and $x_2 = 0.8333$. Since $|x_2| \leq 1$, the model is not invertible.

c) The roots of the quartic equation $\theta(x) = 1 - 0.5x^2 - 0.5x^4 = (x^2 + 2)(0.5 - 0.5x^2)$ are $x_1 = -\sqrt{2}i$, $x_2 = \sqrt{2}i$, $x_3 = -1$ and $x_4 = 1$. Since $|x_3|, |x_4| \leq 1$, the model is not invertible.

Autoregressive Model

Definition 7 (Autoregressive AR(p) Model)

A stochastic process $\{Y_t\}$ follows an AR(p) model if it satisfies

$$\begin{aligned} Y_t &= \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t\,, \text{ or} \\ \phi(B) Y_t &= Z_t\,, \end{aligned}$$

where $Z_t \sim \mathrm{WN}(0, \sigma^2)$ is white noise and
$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$ is the characteristic polynomial.

AR model is very **popular** because

- Closely resembles traditional regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + z$$

  $\therefore$ traditional theoretical results can be used.

- Intuitively, today's value depends on previous days.

AR model with non-zero mean

$\{Y_t\}$ is an AR(p) process with mean μ if it satisfies $\phi(B)(Y_t - \mu) = Z_t$ for all t

$$
\begin{aligned}
(Y_t - \mu) &= \phi_1(Y_{t-1} - \mu) + \cdots + \phi_p(Y_{t-p} - \mu) + Z_t \\
\phi(B)(Y_t - \mu) &= Z_t \\
\phi(B)Y_t &= \phi(1)\mu + Z_t,
\end{aligned}
$$

in which $\phi(1) = 1 - \phi_1 - \phi_2 - \cdots - \phi_p$.

Exercise

Express the AR(2) model $(1 - \phi_1 B - \phi_2 B^2)(Y_t - \mu) = Z_t$ in the form $Y_t = \tilde{\phi}_0 + \tilde{\phi}_1 Y_{t-1} + \tilde{\phi}_2 Y_{t-2} + Z_t$.

Solution:

$$\begin{aligned}(1 - \phi_1 B - \phi_2 B^2)(Y_t - \mu) &= Z_t, \\ Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} - \mu(1 - \phi_1 - \phi_2) &= Z_t,\end{aligned}$$

Hence,

$$Y_t = \mu(1 - \phi_1 - \phi_2) + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t.$$

Exercise

Consider the stationary process $Y_t = \mu + \phi Y_{t-1} + Z_t$, where $Z_t \sim WN(0, \sigma^2)$, and $|\phi| < 1$. Find $\mathrm{E}(Y_t)$ and $\mathrm{Cov}(Y_t, Y_{t-k})$ for $k = 0, 1, 2, \ldots$.

Solution

From the assumed stationarity, we have $\mathrm{E}(Y_t) = \mathrm{E}(Y_{t-1})$, thus $\mathrm{E}(Y_t) = \mu + \phi \mathrm{E}(Y_{t-1}) = \mu + \phi \mathrm{E}(Y_t)$, giving

$$\mathrm{E}(Y_t) = \frac{\mu}{1-\phi}.$$

Taking variance on both sides of $Y_t = \mu + \phi Y_{t-1} + Z_t$, and using the stationarity assumption $\mathrm{Var}(Y_t) = \mathrm{Var}(Y_{t-1})$, we have

$$\begin{aligned} \mathrm{Var}(Y_t) &= \mathrm{Var}(\mu + \phi Y_{t-1} + Z_t) = \phi^2 \mathrm{Var}(Y_{t-1}) + \sigma^2 \\ \Rightarrow \qquad \mathrm{Var}(Y_t) &= \gamma(0) = \frac{\sigma^2}{1-\phi^2}. \end{aligned}$$

Considering the covariance of $Y_t = \mu + \phi Y_{t-1} + Z_t$ with each of Y_{t-k}, $k = 1, 2, \ldots$, we have the autocovariance function $\gamma(k)$:

- $\gamma(1) = \mathrm{Cov}(Y_t, Y_{t-1}) = \mathrm{Cov}(\mu + \phi Y_{t-1} + Z_t, Y_{t-1}) = \phi\gamma(0) = \frac{\phi\sigma^2}{1-\phi^2}$.
- For $k \geq 2$, $\gamma(k) = \mathrm{Cov}(Y_t, Y_{t-k}) = \mathrm{Cov}(\mu + \phi Y_{t-1} + Z_t, Y_{t-k})$
  $= \phi\gamma(k-1) = \phi^2\gamma(k-2) = \cdots = \phi^k\gamma(0) = \frac{\phi^k\sigma^2}{1-\phi^2}$.

Stationarity

Lemma 8 (MA(∞): Stationarity)

If $Y_t = \sum_{i=0}^{\infty} \psi_i Z_{t-i}$, $Z_t \sim WN(0, \sigma^2)$ and $\sum_{i=0}^{\infty} |\psi_i| < \infty$, then $\{Y_t\}$ is a stationary process.

Proof.

Using the properties of white noise, i.e. $\mathrm{E}(Z_t) = 0$, $\mathrm{Cov}(Z_t, Z_k) = \sigma^2 1_{\{t=k\}}$, we have

1. $\mathrm{E}(Y_t) = 0$
2. $\mathrm{Var}(Y_t) = \sigma^2 \sum_{i=0}^{\infty} \psi_i^2 \leq \sigma^2 \left(\sum_{i=0}^{\infty} |\psi_i|\right)^2 < \infty$
3. $\mathrm{Cov}(Y_t, Y_{t+k}) = \sigma^2 \sum_{i=0}^{\infty} \psi_i \psi_{i+k}$ is independent of t

$\Rightarrow$ $\{Y_t\}$ is stationary.

$\square$

AR(1) model: Stationarity

Proposition 9 (Stationarity of AR(1) process)

The process $Y_t = \phi Y_{t-1} + Z_t$, $Z_t \sim WN(0, \sigma^2)$ *is stationary if* $|\phi| \neq 1$.

Proof.

A) For $|\phi| < 1$, inverting the AR polynomial gives
$(1-\phi B)Y_t = Z_t \quad \Longrightarrow \quad Y_t = (1-\phi B)^{-1} Z_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$.

B) For $|\phi| > 1$, inverting a "modified" AR polynomial gives

$$\left(1 - \frac{1}{\phi B}\right)(-\phi B)Y_t = Z_t \quad \Longrightarrow \quad (-\phi B)Y_t = \left(1 - \frac{1}{\phi B}\right)^{-1} Z_t = \sum_{k=0}^{\infty} \frac{1}{\phi^k} Z_{t+k}$$

$$\therefore \; Y_{t-1} = -\sum_{k=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+k} \quad \Longrightarrow \quad Y_t = -\sum_{k=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+1+k}$$

- In both (A) and (B), the coefficients of the WN decay to 0 rapidly as $k \to \infty$ and thus the **summations are well-defined**.
- Both (A) and (B) are stationary (check $\mathrm{E}(Y_t)$ and $\mathrm{Cov}(Y_t, Y_{t+k})$).

  But (B) is not useful: It depends on future noises

AR(1) model with $|\phi| > 1$

Note that when $|\phi| > 1$, the expression

$$Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$$

is not a valid representation since the sum is divergent for $|\phi| \geq 1$. However, if we consider

$$\begin{aligned} Z_t^* &= Y_t - \tfrac{1}{\phi} Y_{t-1} \\ &= Z_t + (\phi - \tfrac{1}{\phi}) Y_{t-1} \\ &= Z_t - (\phi - \tfrac{1}{\phi}) \textstyle\sum_{k=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+k} \\ &= \tfrac{1}{\phi^2} Z_t - (1 - \tfrac{1}{\phi^2}) \textstyle\sum_{k=1}^{\infty} \frac{1}{\phi^k} Z_{t+k}. \end{aligned}$$

It can be shown that Z_t^* defined above is a white noise process with $\mathrm{Var}(Z_t^*) = \frac{\sigma^2}{\phi^2}$.

AR(1) model with $|\phi| > 1$

With the while noise process Z_t^*, Y_t could be regarded as an AR(1) model

$$Y_t = \phi^* Y_{t-1} + Z_t^*$$

with $\phi^* = \frac{1}{\phi}$ such that

$$|\phi^*| = \frac{1}{|\phi|} < 1.$$

Y_t would have a causal representation with respect to Z_t^* as

$$Y_t = \sum_{k=0}^{\infty} \phi^{*k} Z_{t-k}^*.$$

Thus, for a stationary AR(1) model, we typically assume that $|\phi| < 1$.

Example: Stationarity

Example 10

Find a stationary solution for the AR(2) model
$(1-0.3B)(1-1.2B)Y_t = Z_t$.

$$\begin{aligned}
(1-0.3B)(1-1.2B)Y_t &= Z_t \\
\left(1-\tfrac{1}{1.2B}\right)(-1.2B)Y_t &= (1-0.3B)^{-1}Z_t \\
-1.2BY_t &= \left(1-\tfrac{1}{1.2B}\right)^{-1}\textstyle\sum_{k=0}^{\infty} 0.3^k Z_{t-k} \\
Y_{t-1} &= -\tfrac{1}{1.2}\textstyle\sum_{l=0}^{\infty}\tfrac{1}{1.2^l}\textstyle\sum_{k=0}^{\infty} 0.3^k Z_{t+l-k} \\
Y_t &= -\textstyle\sum_{l=0}^{\infty}\textstyle\sum_{k=0}^{\infty}\tfrac{0.3^k}{1.2^{l+1}} Z_{t+1+l-k}
\end{aligned}$$

Theorem 11 (Stationarity of AR(p) process)

The process $\phi(B)Y_t = Z_t$, $Z_t \sim WN(0,\sigma^2)$ is stationary if no root of $\phi(x)$ is on the unit circle.

Exercise

Consider an MA(∞) process $Y_t = Z_t + C(Z_{t-1} + Z_{t-2} + \cdots)$, where C is a fixed constant, $Z_t \sim WN(0, \sigma^2)$.

(a) Is Y_t weakly stationary?

(b) Let $W_t = Y_t - Y_{t-1}$. Show that $\{W_t\}$ is a stationary MA(1) model.

(c) Find the autocorrelation function of $\{W_t\}$.

Solution:

(a) From the definition of Y_t we have

$$\mathrm{Var}(Y_t) = [1 + C^2(1 + 1 + 1 + \ldots)]\sigma^2 = \infty\,,$$

so Y_t is not weakly stationary.

Exercise

(b) Since $W_t = Y_t - Y_{t-1} = Z_t + (C-1)Z_{t-1}$, we have $\mathrm{E}(W_t) = 0$ and autocovariance function $\gamma(k) = \mathrm{Cov}(W_t, W_{t-k})$ satisfying

$$\begin{aligned}
\gamma(0) &= (1+(C-1)^2)\sigma^2 \\
\gamma(1) &= (C-1)\sigma^2 \\
\gamma(k) &= 0 \quad \text{for} \quad |k| > 1\,.
\end{aligned}$$

Thus, $\{W_t\}$ is a stationary MA(1) process.

(c) From the above autocovariance function, we get the autocorrelation function

$$\begin{aligned}
\rho(0) &= 1 \\
\rho(1) &= \frac{C-1}{1+(C-1)^2} \\
\rho(k) &= 0 \quad \text{for} \quad |k| > 1\,.
\end{aligned}$$

Asymptotic stationary

Suppose an AR(1) process does not go back to the remote past, but starts from an initial value Y_0,

$$Y_t = Z_t + \phi Z_{t-1} + \phi^2 Z_{t-2} + \cdots + \phi^{t-1} Z_1 + \phi^t Y_0$$

- Y_0 is usually fixed but **NOT** distributed as $\sum_{k=0}^{\infty} \phi^k Z_{-k}$
- The process Y_t is not stationary because:

$$\begin{aligned} \mathrm{E}(Y_t) &= \phi^t \mathrm{E}(Y_0) \\ \mathrm{Var}(Y_t) &= \sigma^2 \left(1 + \phi^2 + \cdots + \phi^{2(t-1)}\right) + \phi^{2t} \mathrm{Var}(Y_0) \end{aligned}$$

  are time dependent (depends on t).
- However, as $t \to \infty$, $\mathrm{E}(Y_t) \to 0$ and $\mathrm{Var}(Y_t) \to \frac{\sigma^2}{1-\phi^2}$

$\therefore$ we say that $\{Y_t\}$ is asymptotic stationary!

Causal condition

Definition 12 (Causality)

$\{Y_t\}$ is **causal** if it can be expressed as a linear combination of the **past white noise** Z_t, Z_{t-1},...,i.e.,

$$Y_t = \sum_{k=0}^{\infty} \psi_k Z_{t-k} \,,$$

where $\sum_{k=0}^{\infty} |\psi_k| < \infty$ (so that the summation is well defined).

Example 13 (AR(1) Model)

For AR(1) model $Y_t = \phi Y_{t-1} + Z_t$, $Z_t \sim WN(0, \sigma^2)$, we have shown that

- If $|\phi| < 1$, then $Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$

$\Rightarrow$ $\{Y_t\}$ is causal (Note that $\sum_{k=0}^{\infty} |\psi_k| = \sum_{k=0}^{\infty} |\phi^k| = \frac{1}{1-|\phi|} < \infty$)

- If $|\phi| > 1$, then $Y_t = -\sum_{k=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+1+k}$

$\Rightarrow$ $\{Y_t\}$ is **NOT** causal (It depends on future noises)

Remarks on causality and stationarity

What is the difference between stationarity and causality for AR(1) ?

- Causality: depends on past data
  $|\phi| < 1 : Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$ is causal
  $|\phi| > 1 : Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$ no longer exist $\Rightarrow$ **NOT** causal

- Stationarity: constant mean and covariance over time
  - $|\phi| < 1$:
    $Y_t = \sum_{k=0}^{\infty} \phi^k Z_{t-k}$ is a stationary solution
  - $|\phi| > 1$:
    $Y_t = \sum_{i=0}^{\infty} \frac{1}{\phi^{k+1}} Z_{t+1+k}$ is a stationary solution but **NOT** causal

Causality Theorem

Theorem 14 (AR(p) Model)

An AR(p) process $\phi(B)Y_t = Z_t$, $Z_t \sim WN(0, \sigma^2)$ is causal if the roots of the characteristic polynomial

$$\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$$

are outside the unit circle.

Proof.

Factorization: $\phi(B) = (1 - \xi_1 B)(1 - \xi_2 B) \cdots (1 - \xi_p B)$
The roots of $\phi(x)$ are $x = \frac{1}{\xi_1}, \ldots, \frac{1}{\xi_p}$.

- The roots are outside unit circle $\Leftrightarrow$ $\left|\frac{1}{\xi_k}\right| > 1$, for all $k = 1, \ldots, p$

$\Leftrightarrow$ $|\xi_k| < 1$, for all $k = 1, \ldots, p$

$\Leftrightarrow$ $(1 - \xi_k B)$, for all $k = 1, \ldots, p$, can be "inverted" to another side and represented by past noises.

$\Leftrightarrow$ $\{Y_t\}$ is causal. $\square$

Exercise

Which of the following AR model(s) is(are) causal? Give the causal representation up to the second term.

a) $Y_t + 0.6Y_{t-1} - 0.16Y_{t-2} = Z_t$

b) $Y_t - 0.6Y_{t-1} - 0.72Y_{t-2} = Z_t$

c) $Y_t - 0.5Y_{t-2} - 0.5Y_{t-4} = Z_t$

Solution:

a) The roots of the quadratic equation $\phi(x) = 1 + 0.6x - 0.16x^2$ are $x_1 = -1.25$ and $x_2 = 5$. Since $|x_i| > 1$ for $i = 1, 2$, the model is causal.

Solution

To find the causal representation,

$$\begin{aligned} Y_t + 0.6Y_{t-1} - 0.16Y_{t-2} &= Z_t \\ (1+0.8B)(1-0.2B)Y_t &= Z_t. \end{aligned}$$

Hence,

$$\begin{aligned} & Y_t \\ = \ & (1-0.2B)^{-1}(1+0.8B)^{-1}Z_t \\ = \ & \sum_{l=0}^{\infty}\sum_{k=0}^{\infty}(0.2)^l(-0.8)^k Z_{t-k-l} \\ = \ & Z_t - 0.8Z_{t-1} + 0.2Z_{t-1} + (-0.8)^2 Z_{t-2} + 0.2^2 Z_{t-2} - 0.8\cdot 0.2 Z_{t-2} + \dots \\ = \ & Z_t - 0.6Z_{t-1} + 0.52Z_{t-2} + \dots \end{aligned}$$

Solution

b) The roots of the quadratic equation $\phi(x) = 1 - 0.6x - 0.72x^2$ are $x_1 = -1.6667$ and $x_2 = 0.8333$. Since $|x_2| \leq 1$, the model is not causal.

c) The roots of the quartic equation $\theta(x) = 1 - 0.5x^2 - 0.5x^4 = (x^2 + 2)(0.5 - 0.5x^2)$ are $x_1 = -\sqrt{2}i$, $x_2 = \sqrt{2}i$, $x_3 = -1$ and $x_4 = 1$. Since $|x_3|, |x_4| \leq 1$, the model is not causal.

Exercise

Find the AR and MA representations of the following process:

$$Y_t + 0.6Y_{t-1} = Z_t + 0.5Z_{t-1}\,, \quad Z_t \sim WN(0, \sigma^2)\,.$$

Solution

For AR representation, we have

$$
\begin{aligned}
(1+0.6B)Y_t &= (1+0.5B)Z_t \\
\Rightarrow \qquad Z_t &= (1+0.5B)^{-1}(1+0.6B)Y_t \\
&= \sum_{j=0}^{\infty}(-0.5)^j B^j (1+0.6B)Y_t \\
&= \left(\sum_{j=0}^{\infty}(-0.5)^j B^j + 0.6\sum_{j=0}^{\infty}(-0.5)^j B^{j+1}\right) Y_t \\
&= \left(\sum_{j=0}^{\infty}(-0.5)^j B^j + 0.6\sum_{j=1}^{\infty}(-0.5)^{j-1} B^{j}\right) Y_t \\
&= \left(1 + 0.1\sum_{j=1}^{\infty}(-0.5)^{j-1} B^{j}\right) Y_t \\
&= Y_t + 0.1\sum_{j=1}^{\infty}(-0.5)^{j-1} Y_{t-j}\,.
\end{aligned}
$$

Solution

Similarly, we have the MA representation

$$\begin{aligned}
(1+0.6B)Y_t &= (1+0.5B)Z_t \\
\Rightarrow \qquad Y_t &= (1+0.6B)^{-1}(1+0.5B)Z_t \\
&= \sum_{j=0}^{\infty}(-0.6)^j B^j (1+0.5B)Z_t \\
&= \left(\sum_{j=0}^{\infty}(-0.6)^j B^j + 0.5\sum_{j=0}^{\infty}(-0.6)^j B^{j+1}\right) Z_t \\
&= \left(\sum_{j=0}^{\infty}(-0.6)^j B^j + 0.5\sum_{j=1}^{\infty}(-0.6)^{j-1} B^{j}\right) Z_t \\
&= \left(1 - 0.1\sum_{j=1}^{\infty}(-0.6)^{j-1} B^{j}\right) Z_t \\
&= Z_t - 0.1\sum_{j=1}^{\infty}(-0.6)^{j-1} Z_{t-j}\,.
\end{aligned}$$

Exercise

Consider the AR(2) process

$$Y_t = 0.5Y_{t-1} - 0.06Y_{t-2} + Z_t.$$

Find the values of ψ_j, $j = 0, 1, 2, 3...$ if the process is written in the form of general linear model

$$Y_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j} \, .$$

Solution

$$
\begin{aligned}
(1-0.5B+0.06B^2)Y_t &= Z_t \\
\Rightarrow (1-0.2B)(1-0.3B)Y_t &= Z_t \\
\Rightarrow \qquad\qquad\qquad\qquad Y_t &= (1-0.2B)^{-1}(1-0.3B)^{-1}Z_t \\
&= \sum_{j=0}^{\infty}\sum_{k=0}^{\infty} 0.2^j 0.3^k B^{j+k} Z_t \\
&= \sum_{m=0}^{\infty} \sum_{j,k\geq 0, j+k=m} \left(0.2^j 0.3^k\right) B^m Z_t \\
&= \sum_{m=0}^{\infty}\sum_{j=0}^{m} \left(0.2^j 0.3^{m-j}\right) B^m Z_t \\
&= \sum_{m=0}^{\infty} 0.3^m \frac{1-\left(\frac{0.2}{0.3}\right)^{m+1}}{1-\frac{0.2}{0.3}} Z_{t-m}\,.
\end{aligned}
$$

That is, $\psi_j = 0.3^j \frac{1-\left(\frac{0.2}{0.3}\right)^{j+1}}{1-\frac{0.2}{0.3}}$.

Covariance structure of AR(p) model

Q: What is the relationship between the covariance function $\gamma(k)$ and the parameters $\{\phi_i\}$ of a causal AR(p) model $Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t$?

A: Yule-Walker equations can be used

- Multiply both sides by Y_t, then take expectation
- Multiply both sides by Y_{t-1}, then take expectation
- ......
- Multiply both sides by Y_{t-p}, then take expectation

- Solve $p+1$ equations for $p+1$ unknowns for $(\gamma(0), \gamma(1), ..., \gamma(p))$

Yule-Walker equations

$$\text{AR}(p)\text{: } Y_t = \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + Z_t$$

- Multiply both sides by Y_t and take expectation:

$$\begin{aligned} \mathrm{E}(Y_t Y_t) &= \mathrm{E}(\phi_1 Y_{t-1} Y_t + \cdots + \phi_p Y_{t-p} Y_t + Z_t Y_t) \\ \Rightarrow \gamma(0) &= \phi_1 \gamma(1) + \cdots + \phi_p \gamma(p) + \sigma^2. \end{aligned}$$

  Note that stationarity implies that the covariance function is independent of t.

- Multiply both sides by Y_{t-1} and take expectation:

$$\Rightarrow \gamma(1) = \phi_1 \gamma(0) + \phi_2 \gamma(1) + \cdots + \phi_p \gamma(p-1).$$

- In general,

$$\gamma(k) = \phi_1 \gamma(k-1) + \cdots + \phi_p \gamma(k-p), \quad , k = 1, 2, \ldots$$

  Note that causality implies $\mathrm{E}(Y_{t-k} Z_t) = 0$ for $k = 1, 2, \ldots$.

Yule-Walker equations

- We have,

$$\begin{aligned}
\gamma(0) &= \phi_1\gamma(1) + \phi_2\gamma(2) + \cdots + \phi_p\gamma(p) + \sigma^2 \\
\gamma(1) &= \phi_1\gamma(0) + \phi_2\gamma(1) + \cdots + \phi_p\gamma(p-1) \\
\gamma(2) &= \phi_1\gamma(1) + \phi_2\gamma(0) + \cdots + \phi_p\gamma(p-2) \\
\cdots &= \cdots \\
\gamma(p) &= \phi_1\gamma(p-1) + \phi_2\gamma(p-2) + \cdots + \phi_p\gamma(0)\,.
\end{aligned}$$

- In matrix form, we have

$$\begin{pmatrix} 1 & -\phi_1 & -\phi_2 & \cdots & -\phi_p \\ -\phi_1 & 1-\phi_2 & -\phi_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\phi_p & -\phi_{p-1} & -\phi_{p-2} & \cdots & 1 \end{pmatrix} \begin{pmatrix} \gamma(0) \\ \gamma(1) \\ \vdots \\ \gamma(p) \end{pmatrix} = \begin{pmatrix} \sigma^2 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

- We can solve for $\gamma(k)$s once ϕ_js are known

Example: AR(1) Model

Consider a causal AR(1) model,

$$Y_t = \phi Y_{t-1} + Z_t.$$

The Yule-Walker equations yield

$$\begin{aligned}\gamma(0) &= \phi\gamma(1) + \sigma^2, \\ \gamma(k) &= \phi\gamma(k-1), \text{ for } k = 1, 2, \ldots.\end{aligned}$$

The first equation and the second when $k = 1$ implies

$$\gamma(0) = \phi^2\gamma(0) + \sigma^2,$$

and thus $\gamma(0) = \frac{\sigma^2}{1-\phi^2}$. The ACVF is therefore given by

$$\gamma(k) = \frac{\phi^k\sigma^2}{1-\phi^2} \text{ for } k = 0, 1, 2, \ldots.$$

Example: AR(2) Model

Consider a causal AR(2) model,

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t.$$

The Yule-Walker equations yield

$$\begin{aligned}
\gamma(0) &= \phi_1\gamma(1) + \phi_2\gamma(2) + \sigma^2, \\
\gamma(1) &= \phi_1\gamma(0) + \phi_2\gamma(1), \\
\gamma(k) &= \phi_1\gamma(k-1) + \phi_2\gamma(k-2), \text{ for } k = 2, 3, \ldots.
\end{aligned}$$

Divide both sides of the second equation by $\gamma(0)$ yields

$$\rho(1) = \phi_1\rho(0) + \phi_2\rho(1).$$

Since, $\rho(0) = 1$, we have $\rho(1) = \frac{\phi_1}{1-\phi_2}$.

Example: AR(2) Model

Take $k = 2$ in the third equation and divide both sides by $\gamma(0)$ yields

$$\begin{aligned}
\rho(2) &= \phi_1\rho(1) + \phi_2\rho(0) \\
&= \phi_1\frac{\phi_1}{1-\phi_2} + \phi_2 \\
&= \frac{\phi_1^2}{1-\phi_2} + \phi_2.
\end{aligned}$$

Substitute $\gamma(1) = \rho(1)\gamma(0)$ and $\gamma(2) = \rho(2)\gamma(0)$ into the first equation yields

$$\begin{aligned}
\gamma(0) &= \phi_1\rho(1)\gamma(0) + \phi_2\rho(2)\gamma(0) + \sigma^2, \\
\Longrightarrow \gamma(0) &= \frac{\sigma^2}{1-\phi_1\rho(1)-\phi_2\rho(2)}.
\end{aligned}$$

Successive values of $\gamma(k)$ and $\rho(k)$ for $k \geq 3$ can be obtained recursively.

Exercise

Find the ACVF of the following AR models with $Z_t \sim \text{WN}(0,1)$.

a) $Y_t - 0.4Y_{t-1} = Z_t$
b) $Y_t - 0.6Y_{t-1} - 0.27Y_{t-2} = Z_t$
c) $Y_t - 0.6Y_{t-1} - 0.72Y_{t-2} = Z_t$ (Is the answer reasonable? Why?)

Solution:

a) This is an AR(1) model with $\phi = 0.4$ and $\sigma^2 = 1$, we have $\gamma(k) = \frac{0.4^k}{1-0.4^2}$, $k = 0, 1, 2, \ldots$.

Solution

b) This is an AR(2) model with $\phi_1 = 0.6$, $\phi_2 = 0.27$, and $\sigma^2 = 1$, we have

$$
\begin{aligned}
\rho(1) &= \frac{\phi_1}{1-\phi_2} = 0.8219 \\
\rho(2) &= \frac{\phi_1^2}{1-\phi_2} + \phi_2 = 0.7631 \\
\gamma(0) &= \frac{\sigma^2}{1-\phi_1\rho(1)-\phi_2\rho(2)} = 3.3244 \\
\gamma(1) &= \rho(1)\gamma(0) = 2.7325 \\
\gamma(2) &= \rho(2)\gamma(0) = 2.5371 \\
\gamma(3) &= \phi_1\gamma(2) + \phi_2\gamma(1) = 2.2600 \\
\gamma(4) &= \phi_1\gamma(3) + \phi_2\gamma(2) = 2.0410 \\
&\dots
\end{aligned}
$$

Solution

c) This is an AR(2) model with $\phi_1 = 0.6$, $\phi_2 = 0.72$, and $\sigma^2 = 1$, we have

$$\begin{aligned}\rho(1) &= \frac{\phi_1}{1-\phi_2} = 2.1429 \\ \rho(2) &= \frac{\phi_1^2}{1-\phi_2} + \phi_2 = 2.0057\end{aligned}$$

Since $\rho(1)$ and $\rho(2)$ are larger than 1, the results are not reasonable. The problem is that this model is not causal.

Exercise

Consider the process $Y_t = 0.5Y_{t-1} - 0.06Y_{t-2} + Z_t$, $Z_t \sim WN(0,1)$. Find the ACF $\rho(k)$ for $k = 0, 1, 2, 3, 4, ...$

Solution: This is an AR(2) model with $\phi_1 = 0.5$, $\phi_2 = -0.06$ and $\sigma^2 = 1$. We have

$$\rho(0) = 1, \quad \rho(1) = \frac{0.5}{1 - (-0.06)} = 0.472,$$

and for $k = 2, 3, 4, ...$

$$\rho(k) = 0.5\rho(k-1) - 0.06\rho(k-2)\,.$$

Causality condition of AR(2) model

Q: For $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t \Rightarrow \left(1 - \phi_1 B - \phi_2 B^2\right) Y_t = Z_t$,
Any conditions for ϕ_1 and ϕ_2 such that Y_t is causal?

A:
i. Causal
$\Rightarrow$ No root for $\phi(x) = 1 - \phi_1 x - \phi_2 x^2 = 0$ is inside the unit circle
$\Rightarrow$ $\phi(0)$, $\phi(1)$, $\phi(-1)$ are of the same sign
(Otherwise, there exist root(s) between -1 and 1)
$\Rightarrow$ Using $\phi(0) = 1$, $\phi(1) = 1 - \phi_1 - \phi_2$, $\phi(-1) = 1 + \phi_1 - \phi_2$,
we have $\phi_1 + \phi_2 < 1$, $\phi_1 - \phi_2 > -1$.

ii. Causal $\Rightarrow$ Roots for $\phi(x) = 1 - \phi_1 x - \phi_2 x^2 = 0$ are greater than 1
$\Rightarrow$ Roots for $x^2 + \frac{\phi_1}{\phi_2} x - \frac{1}{\phi_2} = 0$ are greater than 1
$\Rightarrow$ |Product of roots| $= |-\frac{1}{\phi_2}|$ is greater than 1
$\Rightarrow$ $|\frac{1}{\phi_2}| > 1$ or $|\phi_2| < 1$

Causality condition of AR(2) model

To conclude, for AR(2): $Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$
The conditions for ϕ_1 and ϕ_2 such that $\{Y_t\}$ is causal are:

1. $\phi_1 + \phi_2 < 1$
2. $\phi_1 - \phi_2 > -1$
3. $|\phi_2| < 1$

Definition of ARMA model

Definition 15

$\{Y_t\}$ is said to be an ARMA(p, q) process if

$$\phi(B)Y_t = \theta(B)Z_t \,, \quad Z_t \sim WN(0, \sigma^2) \,,$$

where

$$\begin{aligned} \phi(B) &= 1 - \phi_1 B - \cdots - \phi_p B^p \\ \theta(B) &= 1 - \theta_1 B - \cdots - \theta_q B^q \end{aligned}$$

are the characteristic polynomials with **NO** common roots.

- If there is a common root for $\phi(B)$ and $\theta(B)$, say $\frac{1}{\xi}$, then,

$$\begin{aligned} \phi(B)Y_t &= \theta(B)Z_t \\ \Rightarrow \cancel{(1 - \xi B)}\phi^*(B)Y_t &= \cancel{(1 - \xi B)}\theta^*(B)Z_t \end{aligned}$$

That is, the ARMA(p, q) reduces to an ARMA$(p - 1, q - 1)$ model.

Why do we use ARMA models?

- In practice, one often needs to use a high order AR or MA to capture the complex structure of the series. Many parameters are needed.
- Sometimes an ARMA model with very few parameters (e.g. ARMA(1,1), ARMA(2,2)) is enough to achieve the goal
- Models with fewer parameters are preferred

Stationarity, Causality and Invertibility of ARMA(p, q) Models

- ARMA(p, q) model: $\phi(B)Y_t = \theta(B)Z_t\,, \quad Z_t \sim WN(0, \sigma^2)\,.$
- Stationary solution exists if
  - All roots of $\phi(B)$ are **not on** the unit circle
- Causal if
  - All roots of $\phi(B)$ are **outside** the unit circle

  $\Rightarrow$ $Y_t = \phi^{-1}(B)\theta(B)Z_t = \sum_{k\geq 0} \psi_k Z_{t-k}$, with $\sum_{k\geq 0} |\psi_k| < \infty$.
- Invertible if
  - All roots of $\theta(B)$ are **outside** the unit circle

  $\Rightarrow$ $Z_t = \theta^{-1}(B)\phi(B)Y_t = \sum_{k\geq 0} \pi_k Y_{t-k}$, with $\sum_{k\geq 0} |\pi_k| < \infty$.

Example of ARMA(p, q) Models

$$\text{ARMA(1,1)}: Y_t - \phi Y_{t-1} = Z_t - \theta Z_{t-1}\,,\, Z_t \overset{\text{iid}}{\sim} N(0, \sigma^2)\,.$$

Q: Is the model causal/invertible? What is the MA representation $Y_t = \sum_{j \geq 0} \psi_j Z_{t-j}$?

A: The model is causal if $|\phi| < 1$, invertible if $|\theta| < 1$.
Given that $|\phi| < 1$, the MA representation is given by

$$\begin{aligned}
&(1 - \phi B)Y_t = (1 - \theta B)Z_t \\
&\Rightarrow Y_t = (1 - \phi B)^{-1}(1 - \theta B)Z_t \\
&\qquad = (1 + \phi B + \phi^2 B^2 + \cdots)(1 - \theta B)Z_t \\
&\qquad = (1 + (\phi - \theta)B + (\phi - \theta)\phi B^2 + (\phi - \theta)\phi^2 B^3 + \cdots)Z_t
\end{aligned}$$

$$\therefore \psi_0 = 1\,,\ \psi_j = (\phi - \theta)\phi^{j-1}\,,\ j = 1, 2, \ldots$$

Q: What is the ACF $\rho(k)$ of the process?

A: **Method 1**: Using MA representation $Y_t = \sum_{j\geq 0} \psi_j Z_{t-j}$, $\psi_0 = 1, \psi_j = (\phi - \theta)\phi^{j-1}$.

$$
\begin{aligned}
\gamma(k) &= \mathrm{Cov}\left(\sum_{j\geq 0} \psi_j Z_{t-j}, \sum_{j\geq 0} \psi_j Z_{t+k-j} \right) = \sum\nolimits_{j=0}^{\infty} \psi_j \psi_{j+k} \sigma^2 \\
&= \begin{cases} \psi_0\psi_k\sigma^2 + \sum_{j=1}^{\infty} \psi_j\psi_{j+k}\sigma^2\,, & k \neq 0 \\ \psi_0^2\sigma^2 + \sum_{j=1}^{\infty} \psi_j^2\sigma^2\,, & k = 0 \end{cases} \\
&= \begin{cases} (\phi-\theta)\phi^{k-1}\sigma^2 + (\phi-\theta)^2 \sum_{j=1}^{\infty} \phi^{2j+k-2}\sigma^2\,, & k \neq 0 \\ \sigma^2 + (\phi-\theta)^2 \sum_{j=1}^{\infty} \phi^{2j-2}\sigma^2\,, & k = 0 \end{cases} \\
&= \begin{cases} (\phi-\theta)\phi^{k-1}\sigma^2 + (\phi-\theta)^2 \frac{\phi^k}{1-\phi^2}\sigma^2\,, & k \neq 0 \\ \sigma^2 + \frac{(\phi-\theta)^2}{1-\phi^2}\sigma^2\,, & k = 0 \end{cases}
\end{aligned}
$$

Q: What is the ACF $\rho(k)$ of the process?

A: **Method 2**: Yule-Walker equation
Multiply Y_t and Y_{t-1} on both sides and take expectation respectively

$$\begin{aligned}
\gamma(0) - \phi\gamma(1) &= (1 - \phi\theta + \theta^2)\sigma^2 \\
\gamma(1) - \phi\gamma(0) &= -\theta\sigma^2
\end{aligned}$$

Solving the two equations gives

$$\gamma(0) = \frac{\left(1 - 2\phi\theta + \theta^2\right)\sigma^2}{1 - \phi^2} \text{ and } \gamma(1) = \frac{(\phi - \theta)\,(1 - \phi\theta)\,\sigma^2}{1 - \phi^2}$$

ARMA(1,1): $Y_t - \phi Y_{t-1} = Z_t - \theta Z_{t-1}\,, Z_t \overset{\text{iid}}{\sim} N(0, \sigma^2)$

For $k > 1$, multiply Y_{t-k} on both sides and take expectation gives

$$\begin{aligned} \gamma(k) - \phi\gamma(k-1) &= 0 \\ \Rightarrow \qquad \gamma(k) &= \phi\gamma(k-1) \\ &= \phi^2\gamma(k-2) \\ &= \cdots \\ &= \phi^{k-1}\gamma(1) \\ &= \phi^{k-1}\frac{(\phi-\theta)(1-\phi\theta)\,\sigma^2}{1-\phi^2}\,, \end{aligned}$$

which agrees with the answer in Method 1.

Exercise

Consider the process

$$Y_t = 0.6Y_{t-1} + Z_t - 0.2Z_{t-1}\,, \quad Z_t \sim WN(0,4)\,.$$

(a) Using Yule-Walker equations, find the ACVF and ACF of $\{Y_t\}$.

(b) Find $\mathrm{Var}(\sum_{t=1}^{4} Y_t)$.

Solution: (a) We have $\phi = 0.6$, $\theta = 0.2$, and $\sigma = 2$. the ACVF is

$$\gamma(k) = \begin{cases} 5 & k = 0\,, \\ 2.2 & k = \pm 1\,, \\ 2.2(0.6)^{|k|-1} & |k| \geq 2\,. \end{cases}$$

The ACF is $\rho(k) = \gamma(k)/\gamma(0)$, which is

$$\rho(k) = \begin{cases} 1 & k = 0\,, \\ 0.44 & k = \pm 1\,, \\ 0.44(0.6)^{|k|-1} & |k| \geq 2\,. \end{cases}$$

Solution

(b) By directly counting the number of autocovaraince terms with different lags, we have

$$\begin{aligned}
\operatorname{Var}\left(\sum_{t=1}^{4} Y_t\right) &= 4\gamma(0) + 6\gamma(1) + 4\gamma(2) + 2\gamma(3) \\
&\qquad \text{(16 terms in total)} \\
&= 4(5) + 6(2.2) + 4(2.2)0.6 + 2(2.2)0.6^2 \\
&= 40.064\,.
\end{aligned}$$

Exercise

Consider the model $Y_t = \theta Y_{t-1} + Z_t + \alpha Z_{t-1} + \beta Z_{t-2}$ where α, β, θ are constants and $Z_t \sim WN(0, \sigma^2)$.

(a) What is the name of this model?

(b) Give the conditions for $\{Y_t\}$ to be (i) stationary, (ii) causal (iii) invertible.

(c) Using (i) $(1-x)^{-1} = \sum_{j=0}^{\infty} x^j$ and (ii) first principle, find the stationary solution of $\{Y_t\}$ for the case $|\theta| < 1$.

(d) Find the stationary solution of $\{Y_t\}$ for the case $|\theta| > 1$.

(e) Let $|\theta| < 1$. Find the variance and autocorrelation function of $\{Y_t\}$.

Solution I

(a) ARMA(1,2) model, given that the characteristic polynomials have no common roots.

(b) (i) The process $\{Y_t\}$ is stationary if $|\theta| \neq 1$. However, if $|\theta| = 1$, Y_t cannot be represented by Z_t with decaying coefficients. Thus Y_t is not stationary.
(ii) The process $\{Y_t\}$ is causal if $\phi(x) \neq 0$ for $|x| \leq 1$, i.e., $|\theta| < 1$.
(iii) The process $\{Y_t\}$ is invertible if $\theta(x) = 1 + \alpha x + \beta x^2 \neq 0$ for $|x| \leq 1$.

Solution II

(c) (i) For $|\theta| < 1$, the stationary solution is given by

$$\begin{aligned}
Y_t &= \theta Y_{t-1} + Z_t + \alpha Z_{t-1} + \beta Z_{t-2} \\
&= (1 - \theta B)^{-1}(1 + \alpha B + \beta B^2) Z_t \\
&= \left(\sum_{j=0}^{\infty} \theta^j B^j \right) (1 + \alpha B + \beta B^2) Z_t \\
&= \left(\sum_{j=0}^{\infty} \theta^j B^j + \sum_{j=0}^{\infty} \alpha \theta^j B^{j+1} + \sum_{j=0}^{\infty} \beta \theta^j B^{j+2} \right) Z_t \\
&= \left(1 + (\theta + \alpha) B + \sum_{j=2}^{\infty} (\theta^2 + \alpha\theta + \beta) \theta^{j-2} B^j \right) Z_t \\
&= Z_t + (\theta + \alpha) Z_{t-1} + \sum_{j=2}^{\infty} (\theta^2 + \alpha\theta + \beta) \theta^{j-2} Z_{t-j} \, .
\end{aligned}$$

Solution III

(ii) Using the first principle, we have

$$\begin{aligned}
Y_t &= \theta Y_{t-1} + Z_t + \alpha Z_{t-1} + \beta Z_{t-2} \\
&= Z_t + (\alpha+\theta) Z_{t-1} + (\beta + \alpha\theta) Z_{t-2} + \beta\theta Z_{t-3} + \theta^2 Y_{t-2} \\
&= Z_t + (\alpha+\theta) Z_{t-1} + (\beta + \alpha\theta + \theta^2) Z_{t-2} + (\alpha\theta^2 + \beta\theta) Z_{t-3} \\
&\quad + \beta\theta^2 Z_{t-4} + \theta^3 Y_{t-3} \\
&= Z_t + (\alpha+\theta) Z_{t-1} + (\beta + \alpha\theta + \theta^2) Z_{t-2} + (\beta + \alpha\theta + \theta^2)\theta Z_{t-3} \\
&\quad + (\beta\theta^2 + \alpha\theta^3) Z_{t-4} + \beta\theta^3 Z_{t-5} + \theta^4 Y_{t-4} \\
&= \cdots \\
&= Z_t + (\theta+\alpha) Z_{t-1} + \sum_{i=2}^{\infty} (\theta^2 + \alpha\theta + \beta)\theta^{i-2} Z_{t-i} \, .
\end{aligned}$$

Solution IV

(d) For $|\theta| > 1$, the stationary solution is given by

$$
\begin{aligned}
Y_t &= \theta Y_{t-1} + Z_t + \alpha Z_{t-1} + \beta Z_{t-2} \\
Y_{t-1} &= -\left(1 - \frac{1}{\theta} B^{-1}\right)^{-1} \frac{1}{\theta}(1 + \alpha B + \beta B^2) Z_t \\
&= -\frac{1}{\theta}\left(\sum_{j=0}^{\infty} \theta^{-j} B^{-j}\right)(1 + \alpha B + \beta B^2) Z_t \\
&= -\frac{1}{\theta}\left(\sum_{j=0}^{\infty} \theta^{-j} B^{-j} + \sum_{j=0}^{\infty} \alpha \theta^{-j} B^{1-j} + \sum_{j=0}^{\infty} \beta \theta^{-j} B^{2-j}\right) Z_t \\
&= -\left(\frac{\beta B^2}{\theta} + \frac{\beta + \theta\alpha}{\theta^2} B + \sum_{j=0}^{\infty} \frac{\beta + \alpha\theta + \theta^2}{\theta^3} \theta^{-j} B^{-j}\right) Z_t \\
Y_t &= -\left(\frac{\beta B^2}{\theta} + \frac{\beta + \theta\alpha}{\theta^2} B + \sum_{j=0}^{\infty} \frac{\beta + \alpha\theta + \theta^2}{\theta^3} \theta^{-j} B^{-j}\right) Z_{t+1} \\
&= -\frac{\beta}{\theta} Z_{t-1} - \frac{\beta + \theta\alpha}{\theta^2} Z_t - \sum_{j=0}^{\infty} \frac{\beta + \alpha\theta + \theta^2}{\theta^3} \theta^{-j} Z_{t+1+j}\,.
\end{aligned}
$$

Solution V

(e) From the result

$$Y_t = Z_t + (\theta + \alpha)Z_{t-1} + \sum_{j=2}^{\infty}(\theta^2 + \alpha\theta + \beta)\theta^{j-2}Z_{t-j}$$

in (a), we have

$$\begin{aligned}\mathrm{Var}(Y_t) &= \left(1 + (\theta+\alpha)^2 + (\theta^2+\alpha\theta+\beta)^2 + \theta^2(\theta^2+\alpha\theta+\beta)^2 + \cdots\right)\sigma^2 \\ &= \sigma^2 + (\theta+\alpha)^2\sigma^2 + \frac{(\theta^2+\alpha\theta+\beta)^2}{1-\theta^2}\sigma^2\,.\end{aligned}$$

The autocovariance function is given by

$$\gamma(k) = \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+|k|}$$

where $\psi_0 = 1$, $\psi_1 = \theta + \alpha$, and $\psi_j = \theta^{j-2}(\theta^2 + \alpha\theta + \beta)$, $j \geq 2$.

Solution VI

In particular,

$$\begin{aligned}\gamma(1) &= \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+1} = \sigma^2 \left(\psi_0 \psi_1 + \psi_1 \psi_2 + \sum_{j=2}^{\infty} \psi_j \psi_{j+1} \right) \\ &= \sigma^2 \left((\theta + \alpha)(\theta^2 + \alpha\theta + \beta + 1) + \frac{\theta(\theta^2 + \alpha\theta + \beta)^2}{1 - \theta^2} \right),\end{aligned}$$

and for $k \geq 2$,

$$\begin{aligned}\gamma(k) &= \sigma^2 \sum_{j=0}^{\infty} \psi_j \psi_{j+k} = \sigma^2 \left(\psi_0 \psi_k + \psi_1 \psi_{k+1} + \sum_{j=2}^{\infty} \psi_j \psi_{j+k} \right) \\ &= \sigma^2 \left((\theta^2 + \alpha\theta + 1)\theta^{k-2}(\theta^2 + \alpha\theta + \beta) + \frac{\theta^k(\theta^2 + \alpha\theta + \beta)^2}{1 - \theta^2} \right).\end{aligned}$$

The autocorrelation function can be obtained by $\rho_k = \gamma(k)/\gamma(0)$. Alternatively, Yule-Walker equations can be used to obtain the same answer.

ARIMA Model

Definition 16

If d times of differencing of a process $\{Y_t\}$, $(1-B)^d Y_t$, is a stationary ARMA(p, q) process, then $\{Y_t\}$ is an ARIMA(p, d, q) process, which satisfies

$$\phi(B)(1-B)^d Y_t = \theta(B) Z_t \,, \quad Z_t \sim WN(0, \sigma^2) \,,$$

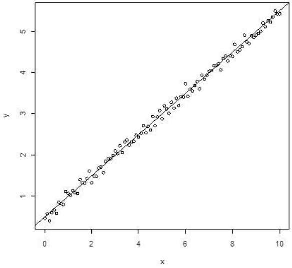
where $\phi(B)$ and $\theta(B)$ have no common roots.

- For $d \geq 1$, the root of $(1-B)^d$ is 1 (on the unit circle), therefore any ARIMA(p, d, q) process is **non-stationary** and **non-causal**.
- But it is **invertible** if the roots of $\theta(B)$ are **outside** the unit circle.

Example 1 of ARIMA Model

Model: $Y_t = \alpha + \beta t + N_t$

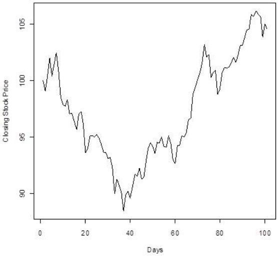
- $\{Y_t\}$ is non-stationary (mean level changes with time)
- $$(1-B)Y_t = \underbrace{\beta + N_t - N_{t-1}}_{\text{MA(1) with unit root}}$$
- $Y_t \sim$ ARIMA(0,1,1)
- Y_t is non-stationary, non-causal, non-invertible
- $(1-B)Y_t$ is stationary, causal but non-invertible



Example 2 of ARIMA Model

Model (Random Walk): $Y_t = Y_{t-1} + Z_t$

- Expressed as $(1-B)Y_t = Z_t$
- $\{Y_t\} \sim$ ARIMA(0,1,0)
- $\{Y_t\}$ is non-stationary: If $Y_0 = 0, Y_t = Z_t + Z_{t-1} + \cdots + Z_1$ then $\mathrm{Var}(Y_t) = t\sigma^2$ changes with time!



What does 'integrated' mean?

- Let $\{Y_t\}$ be an ARIMA(p,1,q) process, then **differencing** the ARIMA Y_t gives an ARMA $W_t = (1 - B)Y_t$.
- In other words, the ARIMA Y_t is obtained from summing **(integrating)** the ARMA W_t:

$$\sum_{k=1}^{t} W_k = \sum_{k=1}^{t}(Y_k - Y_{k-1}) = Y_t - Y_0 = Y_t \quad \text{if} \quad Y_0 = 0$$

Stock Price and ARMA model

Let $P_t =$ Price of stock at the end of date t and
$Y_t = \log P_t = \log(\text{stock price})$

- By Taylor's series expansion of $f(x) = \log x$ around $x = 1$,

$$\log x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 + \cdots$$

$$\begin{aligned} Y_t - Y_{t-1} &= \log \tfrac{P_t}{P_{t-1}} \\ &= \tfrac{P_t}{P_{t-1}} - 1 + \text{ smaller terms} \\ &\approx \tfrac{P_t - P_{t-1}}{P_{t-1}} = \text{daily return} \triangleq Z_t \end{aligned}$$

- Common Assumption:

$\log P_t = Y_t$ follow ARIMA(0,1,0) (Random Walk)

Seasonal ARIMA

$$\text{Seasonal Component:}\, Y_t \approx Y_{t-s} \approx Y_{t-2s} \approx Y_{t-3s} \approx \cdots$$

Example

$$Y_t = \phi_1 Y_{t-4} + \phi_2 Y_{t-8} + Z_t$$

can be used to model quarterly data
(e.g. this summer depends on last summer and the summer before the last one ...)

Seasonal ARIMA

Definition 17

$\{Y_t\}$ follows an SARIMA$(p, d, q) \times (P, D, Q)_s$ model if

$$\phi(B)\Phi_P(B^s)(1-B)^d(1-B^s)^D Y_t = \theta(B)\Theta_Q(B^s)Z_t$$

where

$$\begin{aligned}
\phi(B) &= 1 - \phi_1 B - \cdots - \phi_p B^p \\
\theta(B) &= 1 - \theta_1 B - \cdots - \theta_q B^q \\
\Phi_P(B^s) &= 1 - \Phi_1 B^s - \cdots - \Phi_P B^{sP} \\
\Theta_Q(B^s) &= 1 - \Theta_1 B^s - \cdots - \Theta_Q B^{sQ}
\end{aligned}$$

Remarks:

- SARIMA can be expressed as high order ARIMA model
- But we love the physical interpretation about seasonal effect

Exercise

Identify the order of the following Seasonal-ARIMA models

a) $Y_t - 0.2Y_{t-1} - 0.3Y_{t-4} + 0.06Y_{t-5} = Z_t + 0.6Z_{t-4} - 0.16Z_{t-8}$

b) $Y_t + 0.2Y_{t-1} - Y_{t-6} - 0.2Y_{t-7} = Z_t + 0.2Z_{t-6}$

c) $Y_t - Y_{t-1} + 0.2Y_{t-6} - 0.2Y_{t-7} = Z_t + 0.2Z_{t-6}$

Solution:

a)

$$\begin{aligned}
Y_t - 0.2Y_{t-1} - 0.3Y_{t-4} + 0.06Y_{t-5} &= Z_t + 0.6Z_{t-4} - 0.16Z_{t-8} \\
(1 - 0.2B - 0.3B^4 + 0.06B^5)Y_t &= (1 + 0.6B^4 - 0.16B^8)Z_t \\
(1 - 0.2B)(1 - 0.3B^4)Y_t &= (1 - 0.2B^4)(1 + 0.8B^4)Z_t.
\end{aligned}$$

It is a SARIMA$(1,0,0) \times (1,0,2)_4$ model.

Solution

b)

$$
\begin{aligned}
Y_t + 0.2Y_{t-1} - Y_{t-6} - 0.2Y_{t-7} &= Z_t + 0.2Z_{t-6} \\
(1 + 0.2B - B^6 - 0.2B^7)Y_t &= (1 + 0.2B^6)Z_t \\
(1 + 0.2B)(1 - B^6)Y_t &= (1 + 0.2B^6)Z_t.
\end{aligned}
$$

It is a SARIMA$(1, 0, 0) \times (0, 1, 1)_6$ model.

c)

$$
\begin{aligned}
Y_t - Y_{t-1} + 0.2Y_{t-6} - 0.2Y_{t-7} &= Z_t + 0.2Z_{t-6} \\
(1 - B + 0.2B^6 - 0.2B^7)Y_t &= (1 + 0.2B^6)Z_t \\
(1 - B)(1 + 0.2B^6)Y_t &= (1 + 0.2B^6)Z_t \\
(1 - B)Y_t &= Z_t
\end{aligned}
$$

It is an ARIMA$(0, 1, 0)$ model.

Summary

- **ARIMA model**: $\phi(B)(1-B)^d Y_t = \theta(B)Z_t$, $Z_t \sim WN(0, \sigma^2)$

- **Stationarity, Causality and Invertibility** -
  Characteristic roots of $\phi(x) = 0$, $\theta(x) = 0$ lie outside unit circle

- Find **Causal/MA representation** $Y_t = \sum_{j \geq 0} \psi_j Z_{t-j}$

  Find **AR representation** $Z_t = \sum_{j \geq 0} \psi_j Y_{t-j}$

- Find **ACF** $\gamma(k)$
  (M1) Use causal representation $Y_t = \sum_{j \geq 0} \psi_j Z_{t-j}$
  (M2) Multiply both sides by Y_{t-k}, $k =$ 0,1,...,p and take expectation, then solve for $\gamma(k)$s.

- **SARIMA model**:
  $\phi(B)\Phi_P(B^s)(1-B)^d(1-B^s)^D Y_t = \theta(B)\Theta_Q(B^s)Z_t$